

THE SECANT METHOD AND OTHER QUASI-NEWTON METHODS

Lecture 16: Presented by Jan van Vuuren



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The Secant method (p. 78)

A disadvantage of Newton's method, despite its impressive convergence rate, is that its accuracy comes at the computational cost of having to compute second-order derivatives of the function being optimised

In this lecture, we consider the so-called *secant method* for approximating roots of continuous functions without having to compute their derivatives

The word *secant* derives from the late 16th century French word *sécante*

This word is, in turn, based on the Latin *secare*, meaning *to cut*

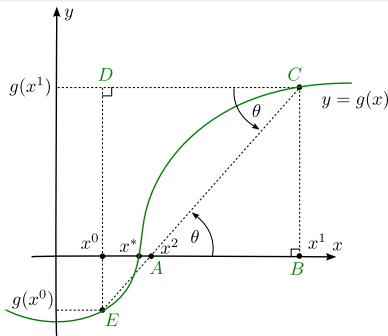
In mathematics a *secant* is a line that cuts a curve into two or more pieces

Instead, the working of the secant method is based on an angle formed by drawing a straight line cutting through two points on the curve

The method may be thought of as a finite-difference approximation of Newton's method, but predates Newton's method by more than 3 000 years

The Secant method generates a sequence $x^2, x^3, x^4, \dots$ approximations of the root of a function $g : \mathbb{R} \mapsto \mathbb{R}$, starting out from a pair of initial estimates x^0 and x^1

The Secant method (pp. 78–79)



From $\triangle ABC$ and $\triangle CDE$: $\tan \theta = \frac{g(x^1)}{x^1 - x^2} = \frac{g(x^1) - g(x^0)}{x^1 - x^0},$

$\Downarrow$

$$x^2 = x^1 - \frac{x^0 - x^1}{g(x^0) - g(x^1)} g(x^1)$$

$\Downarrow$

Generalising: $x^{k+1} = x^k - \frac{x^{k-1} - x^k}{g(x^{k-1}) - g(x^k)} g(x^k), \quad k = 1, 2, 3, \dots$

Secant method (for roots) in pseudo-code (p. 79)

Algorithm 3.8 (Secant method for univariate function root approximation)

Input: An instance of a univariate function $g : \mathbb{R} \mapsto \mathbb{R}$, a pair of initial estimates x^0 and x^1 as to a root of g , and a search termination criterion

Output: An approximation of a root of f satisfying the search termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search criterion is met, then output x^k and stop

Step 3 $x^{k+1} \leftarrow x^k + \Delta(x^k)$, where $\Delta(x^k) = -\frac{x^{k-1} - x^k}{g(x^{k-1}) - g(x^k)} g(x^k)$

Step 4 $k \leftarrow k + 1$

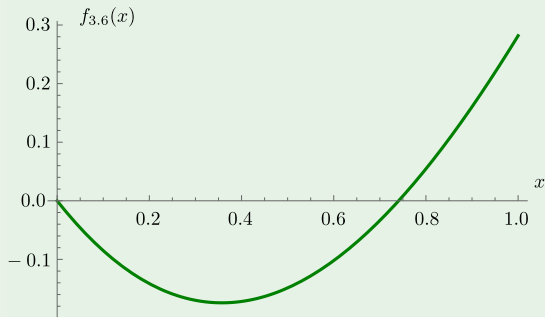
Step 5 Return to Step 2

Theorem 3.5 (Convergence rate of the Secant method)

Let $g : \mathbb{R} \mapsto \mathbb{R}$ be a twice continuously differentiable function and let x^* be a root of g with multiplicity one such that $g'(x^*) \neq 0$. If x^0 and x^1 are sufficiently close to x^* , then the sequence $x^0, x^1, x^2, \dots$ of search points generated by Algorithm 3.8 converges super-linearly to x^* . The convergence rate is $r = (1 + \sqrt{5})/2$ and the rate constant is $C = |g''(x^*)|/2|g'(x^*)|$.

Example 3.13 (pp. 79–80)

Consider again the function $f_{3.6}(x) = 1 + 2x^2 - e^x$



We apply the secant method to approximate the positive root of $g = f_{3.6}$

We start out from the initial estimates $x^0 = 0.8$ and $x^1 = 0.9$

Example 3.13 (pp. 79–80)

$$x^{k+1} \leftarrow x^k + \Delta(x^k), \text{ where } \Delta(x^k) = -\frac{x^{k-1} - x^k}{g(x^{k-1}) - g(x^k)} g(x^k)$$

We work accurately to 6 decimal places and terminate the search once successive estimates no longer change in the first 3 decimal places, *i.e.* if $|x^{k+1} - x^k|$ drops below 0.000 5

k	x^{k-1}	x^k	$f_{3.6}(x^{k-1})$	$f_{3.6}(x^k)$	$\Delta(x^k)$
1	0.8	0.9	0.054 459	0.160 397	−0.151 407
2	0.9	0.748 593	0.160 397	0.006 760	−0.006 662
3	0.748 593	0.741 932	0.006 760	0.000 937	−0.001 072
4	0.741 932	0.740 860	0.000 937	0.000 008	−0.000 009
5	0.740 860	0.740 851	—	—	—

It follows that $x^* \approx \mathbf{0.740\,851}$ (accurately to at least 3 decimal places)

Approximating optima rather than roots (p. 80)

Suppose, however, we sought not the root of a function, but instead a local optimum of the function

Recall that Theorem 2.4 guarantees that $f'(x^*) = 0$ at any local optimum x^* of a univariate function $f(x) : \mathbb{R} \mapsto \mathbb{R}$

We may therefore substitute the derivative f' for the function g into (3.29) to arrive at the formula

$$x^{k+1} = x^k - \frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)} f'(x^k), \quad k = 1, 2, 3, \dots$$

for generating a sequence $x^2, x^3, x^4, \dots$ of approximations of a local optimum x^* of f , starting out from a pair of initial estimates x^0 and x^1

Secant method (optimisation) in pseudo-code (p. 80)

Algorithm 3.9 (Secant method for univariate function optimisation)

Input: An instance of a univariate function $f : \mathbb{R} \mapsto \mathbb{R}$, a pair of initial estimates x^0 and x^1 as to a local optimum of f , and a search termination criterion

Output: An approximation of a local optimum of f satisfying the search termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search criterion is met, then output x^k and stop

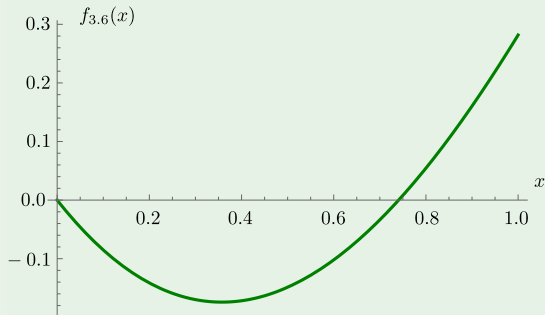
Step 3 $x^{k+1} \leftarrow x^k + \Delta(x^k)$, where $\Delta(x^k) = -\frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)} f'(x^k)$

Step 4 $k \leftarrow k + 1$

Step 5 Return to Step 2

Example 3.14 (p. 81)

Consider again the function $f_{3.6}(x) = 1 + 2x^2 - e^x$



We apply the Secant method to approximate the minimum x^* of $g = f_{3.6}$ by approximating the root of $f(x) = f_{3.7}(x) = f'_{3.6}(x) = 4x - e^x$

We start out from the initial estimates $x^0 = 0.4$ and $x^1 = 0.5$

Example 3.14 (p. 81)

$$x^{k+1} \leftarrow x^k + \Delta(x^k), \text{ where } \Delta(x^k) = -\frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)} f'(x^k)$$

We work accurately to 6 decimal places and terminate the search once successive estimates no longer change in the first 3 decimal places, *i.e.* if $|x^{k+1} - x^k|$ drops below 0.000 5

k	x^{k-1}	x^k	$f_{3.7}(x^{k-1})$	$f_{3.7}(x^k)$	$\Delta(x^k)$
1	0.4	0.5	0.108 175	0.351 279	-0.144 498
2	0.5	0.355 502	0.351 279	-0.004 888	0.001 983
3	0.355 502	0.357 485	-0.004 888	0.000 212	-0.000 082
4	0.357 485	0.357 403	—	—	—

It follows that $x^* \approx \mathbf{0.357\,403}$ (accurately to at least 3 decimal places)

Quasi-Newton methods (pp. 81–82)

Newton's method:

$$x^{k+1} = x^k - \frac{1}{f''(x^k)} f'(x^k)$$

The Secant method:

$$x^{k+1} = x^k - \frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)} f'(x^k)$$

The Secant method may be thought of as approximating Newton's method, with the reciprocal of second-order derivative $1/f''(x^k)$ in Newton's method having been replaced by the term

$$\frac{x^{k-1} - x^k}{f'(x^{k-1}) - f'(x^k)}$$

in the Secant method

That is, the Secant method resembles Newton's method, or is *quasi-Newton*, by virtue of the approximation

$$f''(x^k) \approx \frac{f'(x^k) - f'(x^{k-1})}{x^k - x^{k-1}}$$

$$\begin{aligned} f''(x^k) &\approx \frac{f'(x^k) - f'(x^{k-1})}{x^k - x^{k-1}} \\ &\Downarrow \\ f''(x^k)(x^k - x^{k-1}) &\approx f'(x^k) - f'(x^{k-1}) \end{aligned}$$

Generalising to higher dimensions:

$$\begin{aligned} \nabla^2 f(\mathbf{x}^k)(\mathbf{x}^k - \mathbf{x}^{k-1}) &\approx \nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1}) \\ &\Downarrow \\ \mathbf{x}^k &\approx \mathbf{x}^{k-1} + [\nabla^2 f(\mathbf{x}^k)]^{-1}[\nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})] \end{aligned}$$

Or, for some approximation $\mathbf{B}^k$ of the Hessian matrix $\nabla^2 f(\mathbf{x}^k)$,

$$\mathbf{B}^k(\mathbf{x}^k - \mathbf{x}^{k-1}) = \nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})$$

which may be simplified to

$$\mathbf{B}^k \mathbf{s}^{k-1} = \mathbf{y}^{k-1}$$

where $\mathbf{s}^{k-1} = \mathbf{x}^k - \mathbf{x}^{k-1}$ and $\mathbf{y}^{k-1} = \nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})$

Secant condition:

$$\mathbf{B}^k \mathbf{s}^{k-1} = \mathbf{y}^{k-1}$$

where $\mathbf{s}^{k-1} = \mathbf{x}^k - \mathbf{x}^{k-1}$ and $\mathbf{y}^{k-1} = \nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})$

$\Downarrow$

$$\mathbf{B}^{k+1} \mathbf{s}^k = \mathbf{y}^k$$

where $\mathbf{s}^k = \mathbf{x}^{k+1} - \mathbf{x}^k$ and $\mathbf{y}^k = \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

The update strategy during iteration $k + 1$ of a quasi-Newton method is

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha^k \mathbf{p}^k,$$

where $\mathbf{p}^k$ is the search direction during the k -th iteration, which satisfies

$$\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$$

The Hessian approximation $\mathbf{B}^k$ is updated according to the formula

$$\mathbf{B}^{k+1} = \mathbf{B}^k + \Delta^k$$

where Δ^k is a correction term that differs from one quasi-Newton method to the next

Quasi-Newton methods (pp. 81–82)

Algorithm 3.10 (Any quasi-Newton method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian estimate $\mathbf{B}^0$ as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k$ by Gauss-Jordan elimination to solve the system of linear equations $\mathbf{B}^k \mathbf{p}^k = -\nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

Step 8 $\mathbf{B}^{k+1} \leftarrow \mathbf{B}^k + \Delta^k$ (where Δ^k is computed according to an appropriate update formula)

Step 9 $k \leftarrow k + 1$ and return to Step 2

Returning to (3.32),

$$\mathbf{x}^k - \mathbf{x}^{k-1} \approx [\nabla^2 f(\mathbf{x}^k)]^{-1} [\nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})]$$

in which we could have chosen to estimate the Hessian matrix inverse directly by a matrix $\mathbf{H}^k$, in which case we would have obtained the alternative *secant condition*

$$\mathbf{x}^k - \mathbf{x}^{k-1} = \mathbf{H}^k [\nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})]$$

or

$$\mathbf{s}^{k-1} = \mathbf{H}^k \mathbf{y}^{k-1},$$

where $\mathbf{s}^{k-1} = \mathbf{x}^k - \mathbf{x}^{k-1}$ and $\mathbf{y}^{k-1} = \nabla f(\mathbf{x}^k) - \nabla f(\mathbf{x}^{k-1})$, as before

This would have lead to the update strategy

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha^k \mathbf{p}^k,$$

where $\mathbf{p}^k$ is again the search direction during the k -th iteration, given by

$$\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$$

The Hessian inverse approximation $\mathbf{H}^k$ would then be updated according to the formula

$$\mathbf{H}^{k+1} = \mathbf{H}^k + \tilde{\Delta}^k$$

Algorithm 3.10 (Any quasi-Newton method)

Input: An instance of a multivariate differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ as to an approximate local optimum of f , an initial Hessian inverse estimate $\mathbf{H}^0$ as well as a search termination criterion

Output: An approximation of a local optimum of f close to $\mathbf{x}^0$ satisfying the termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Compute the search direction $\mathbf{p}^k = -\mathbf{H}^k \nabla f(\mathbf{x}^k)$

Step 4 Compute an appropriate step length α^k in the search direction $\mathbf{p}^k$ by performing an exact line search or a backtracking Armijo-Goldstein line search in respect of the function $f(\mathbf{x}^k + \alpha^k \mathbf{p}^k)$

Step 5 Compute the next search estimate as $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{p}^k$

Step 6 $\mathbf{s}^k \leftarrow \mathbf{x}^{k+1} - \mathbf{x}^k$

Step 7 $\mathbf{y}^k \leftarrow \nabla f(\mathbf{x}^{k+1}) - \nabla f(\mathbf{x}^k)$

Step 8 $\mathbf{H}^{k+1} \leftarrow \mathbf{H}^k + \tilde{\Delta}^k$ (where $\tilde{\Delta}^k$ is computed according to an appropriate update formula)

Step 9 $k \leftarrow k + 1$ and return to Step 2

Algorithm 3.10 is implemented by approximating a sequence

$$B^0, B^1, B^2, \dots$$

of Hessian matrix estimates or else approximating a sequence

$$H^0, H^1, H^2, \dots$$

of Hessian inverse matrix estimates (not a mixture between the two)

In the four lectures that follow, we consider arguably the four most popular members of the class of quasi-Newton methods, namely:

- Broyden's method,
- the symmetric rank-one method,
- the Broyden-Fletcher-Goldfarb-Shanno method, and
- the Davidon-Fletcher-Powell method

Each of these methods has a different correction term Δ^k or $\tilde{\Delta}^k$ in Step 8 of Algorithm 3.10